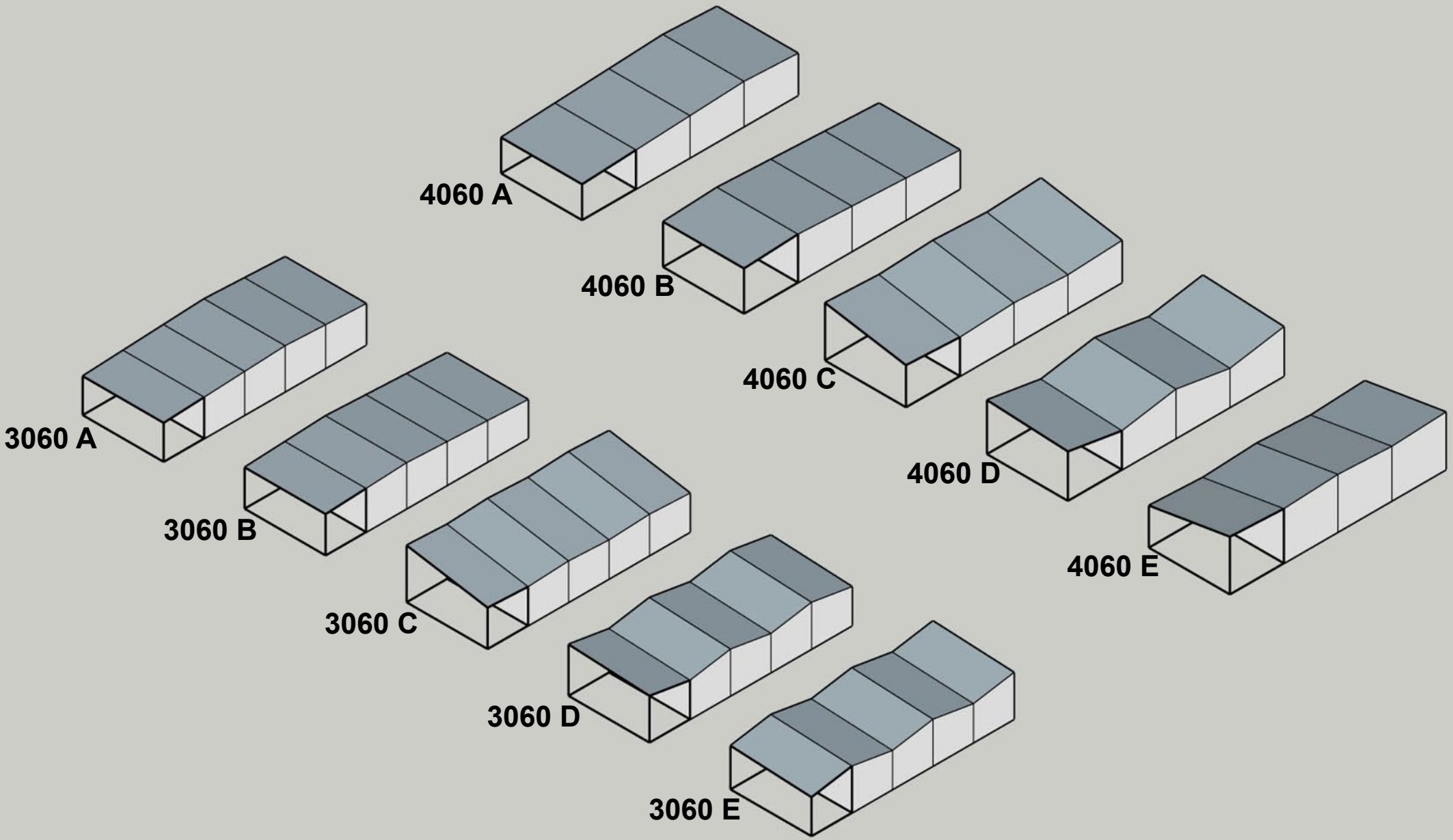
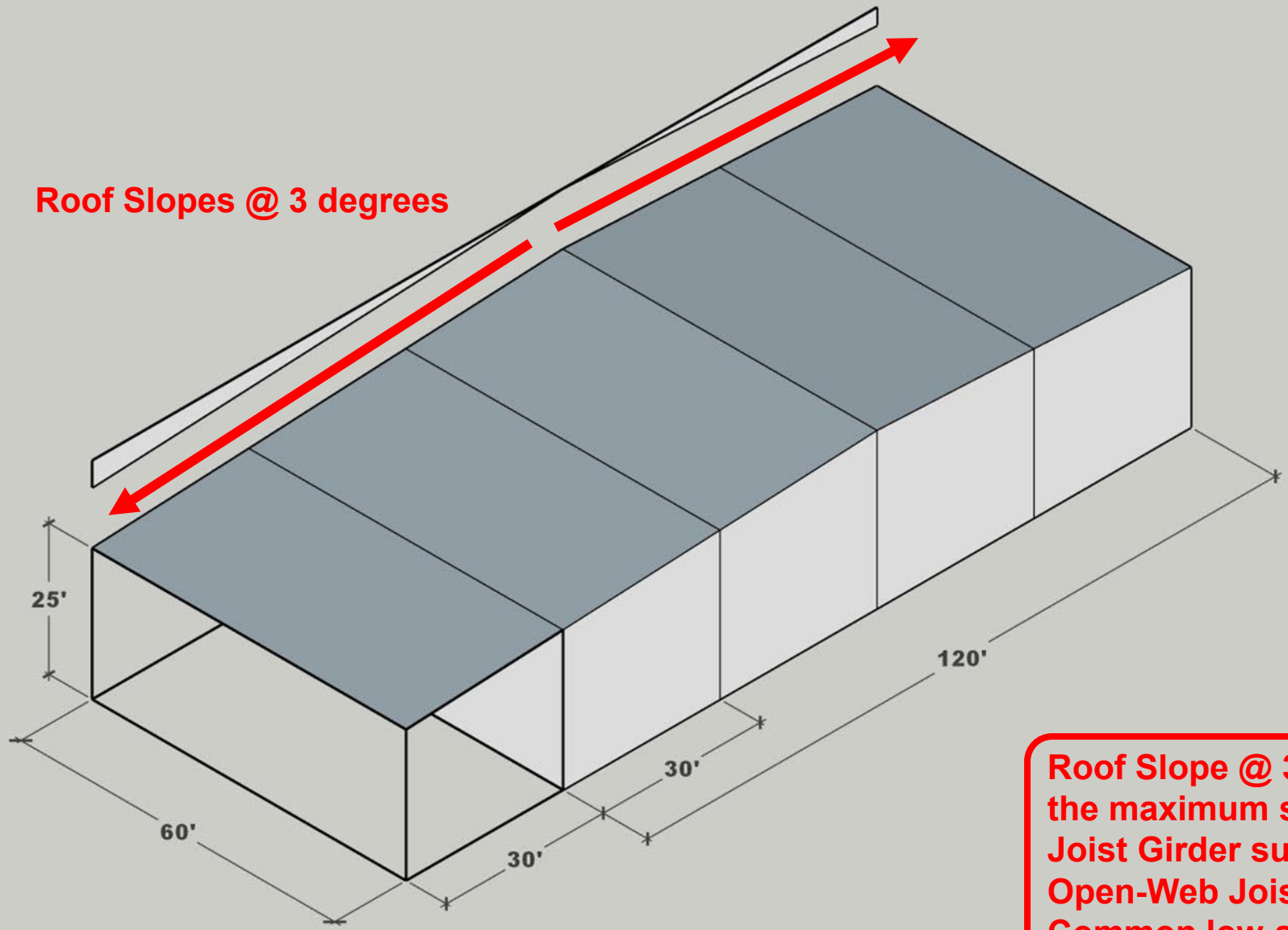
STRUCTURAL SYSTEMS 1 *ARC 2713*

STEEL BUILDING: LAYOUT OPTIONS FOR ROOF & FLOOR + PROJECT PARAMETERS

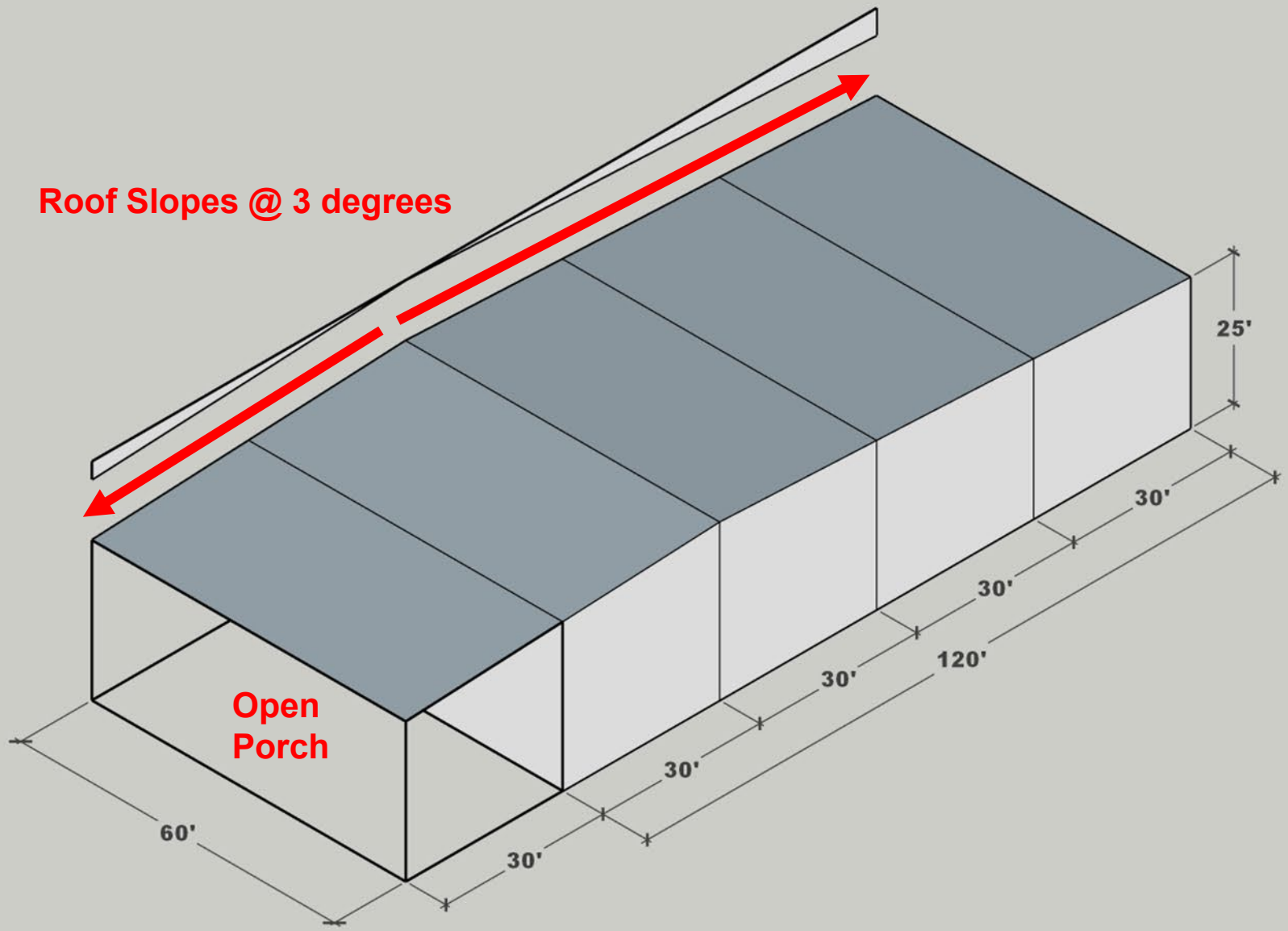


ROOF LAYOUT OPTIONS: 30 x 60 bays or 40 x 60 bays, each with an open “Porch” at one end

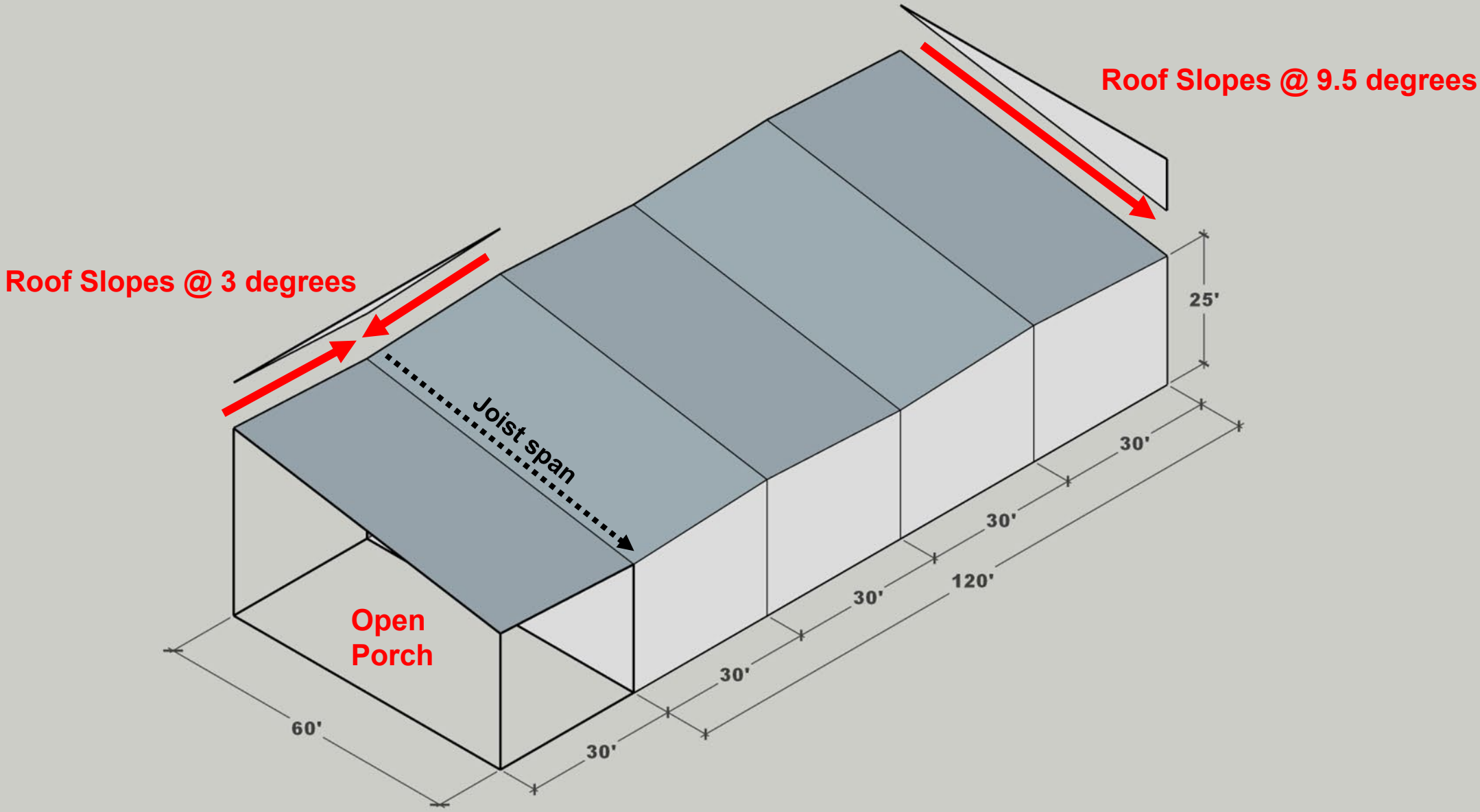


Roof Slope @ 3 degrees is the maximum slope for a Joist Girder supporting Open-Web Joists. Common low-slope for metal roofing or membrane roofing is 3 degrees .

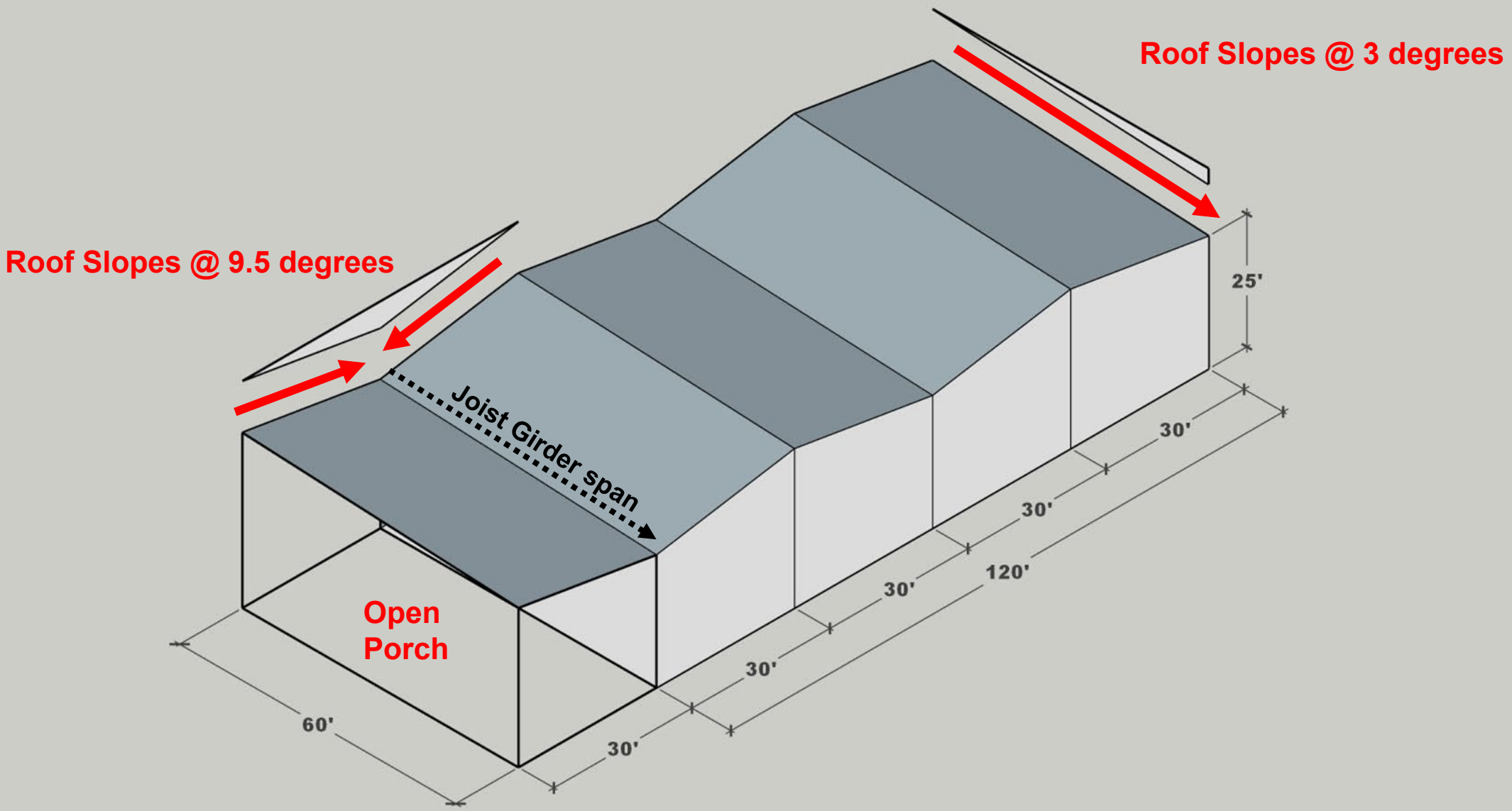
3060 A Roof Slope



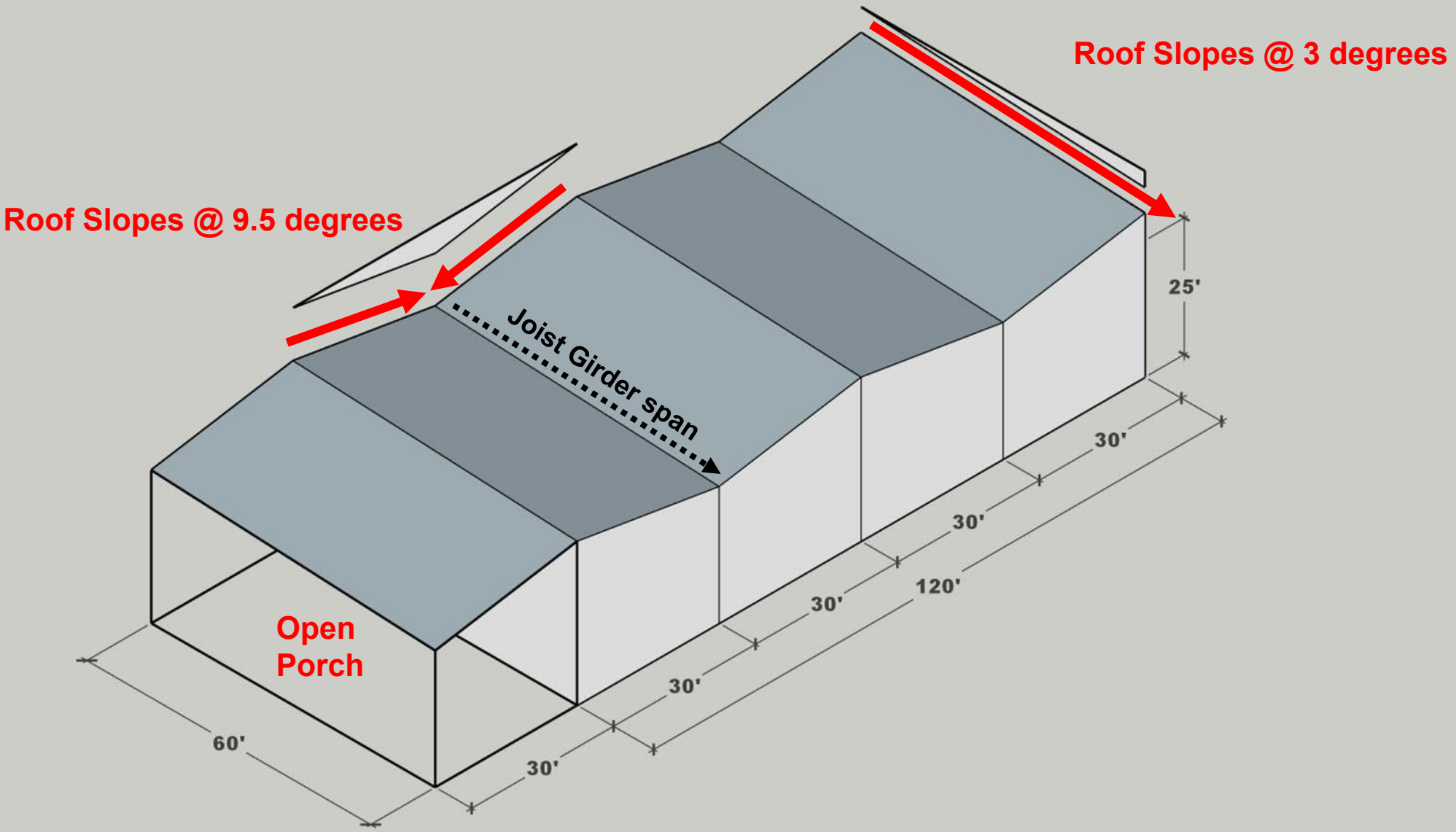
3060 B (30’ x 60’ structural roof bay) **(Joists and Joist Girders have 2 options** like 3060 A)



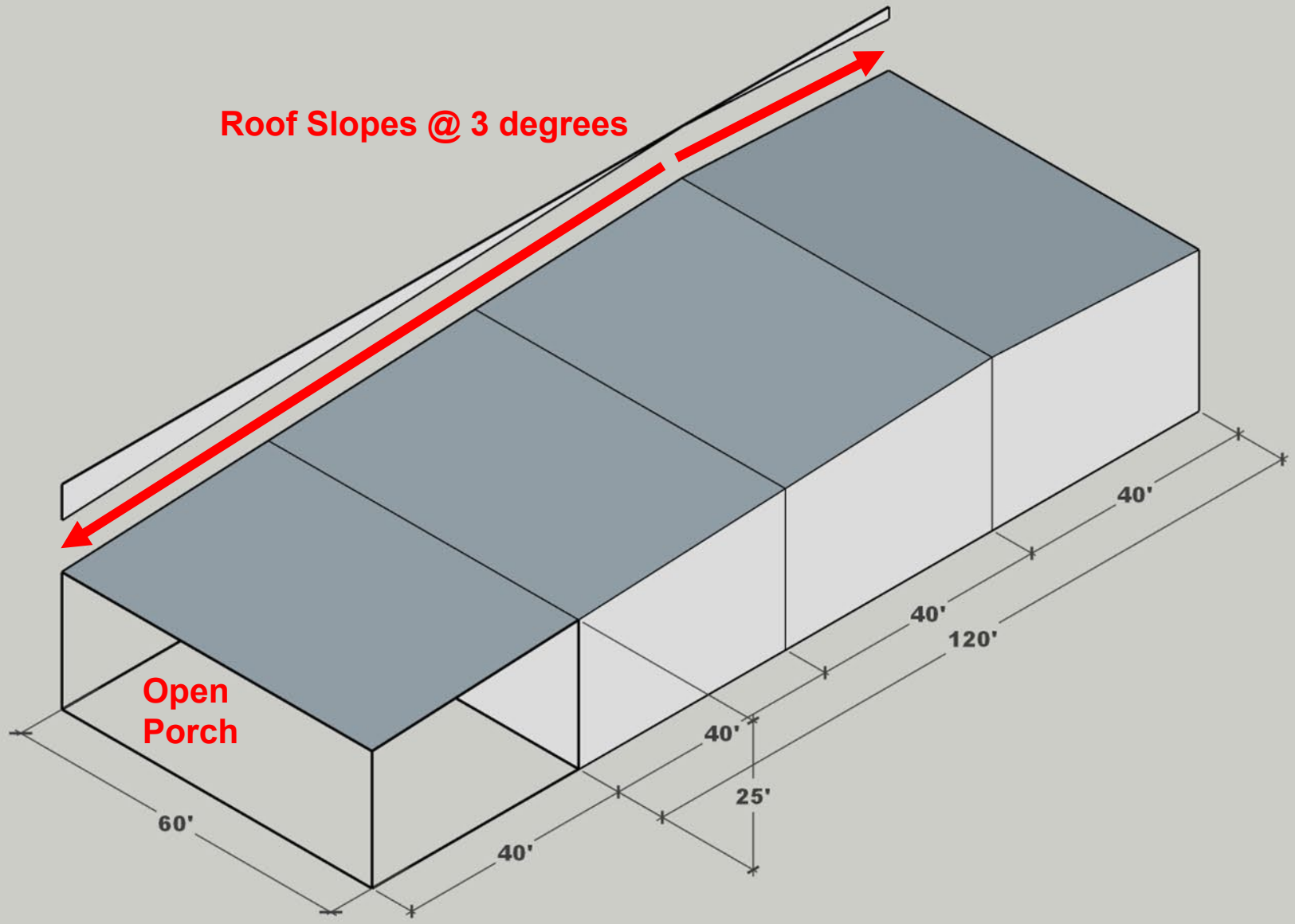
3060 C (30’ x 60’ structural roof bay) **(Joists <u>must</u> span 60’)**



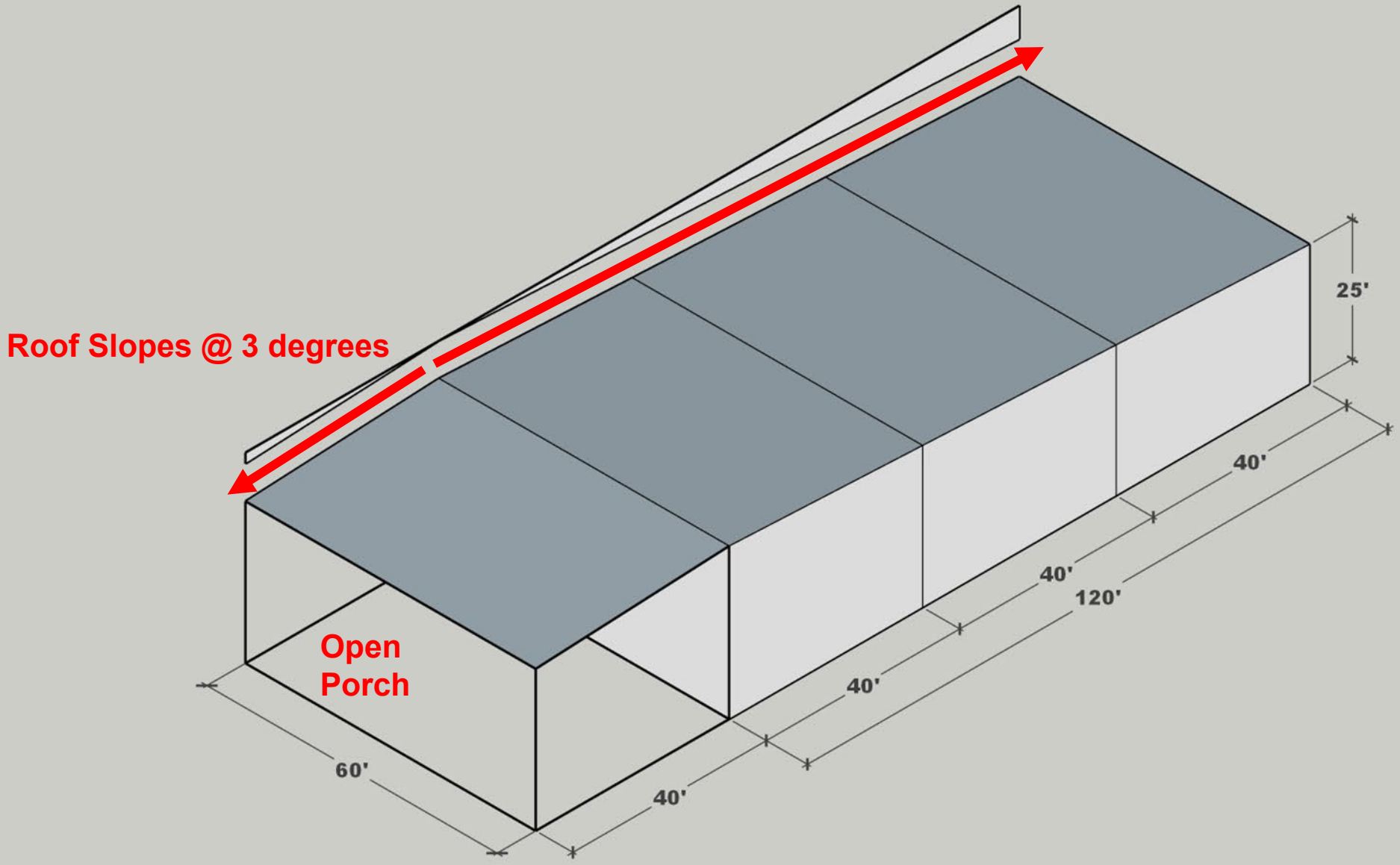
3060 D (30’ x 60’ structural roof bay) **(Joist Girders must span 60’)**



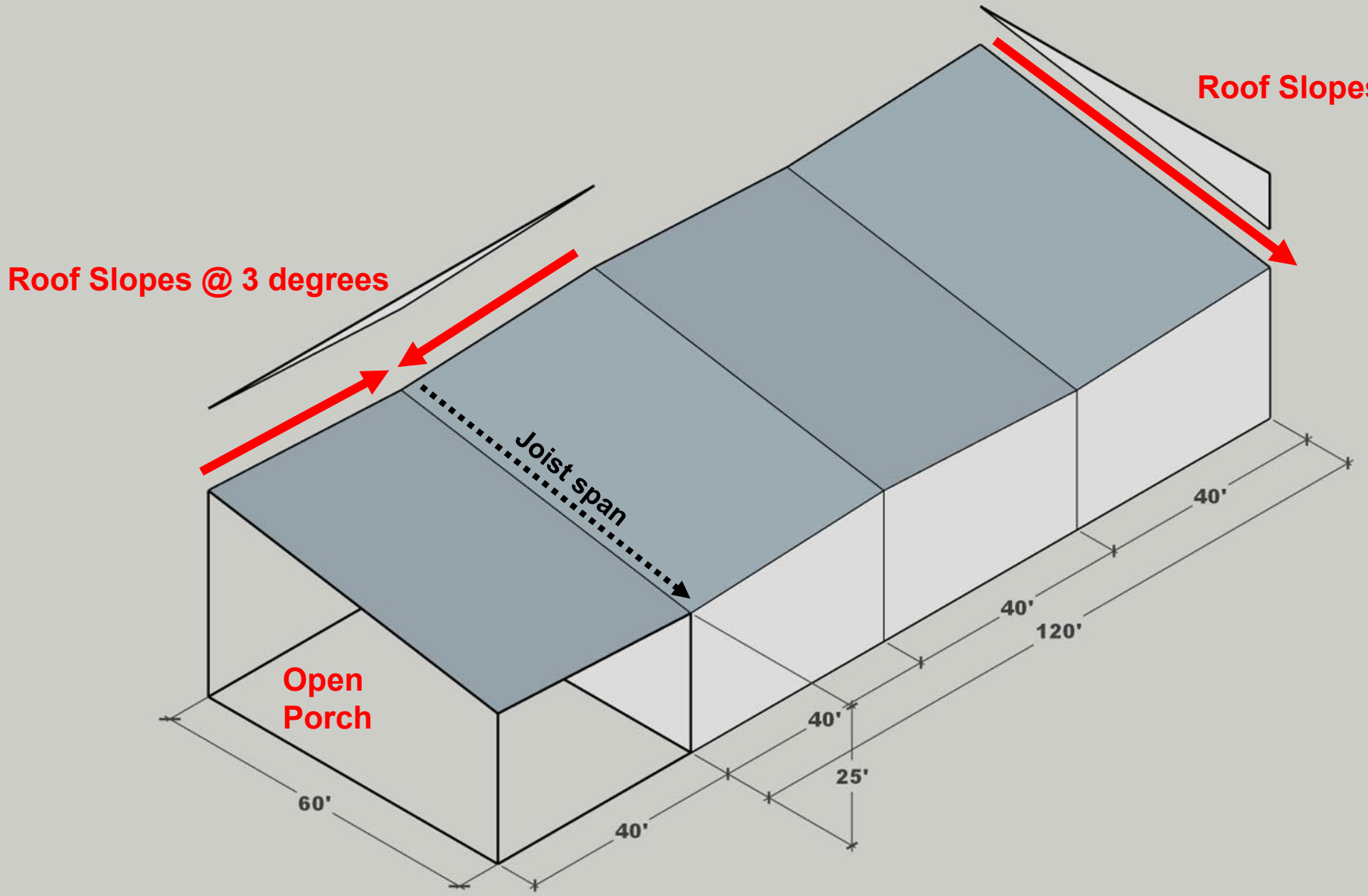
3060 E (30’ x 60’ structural roof bay) **(Joist Girders must span 60’)**



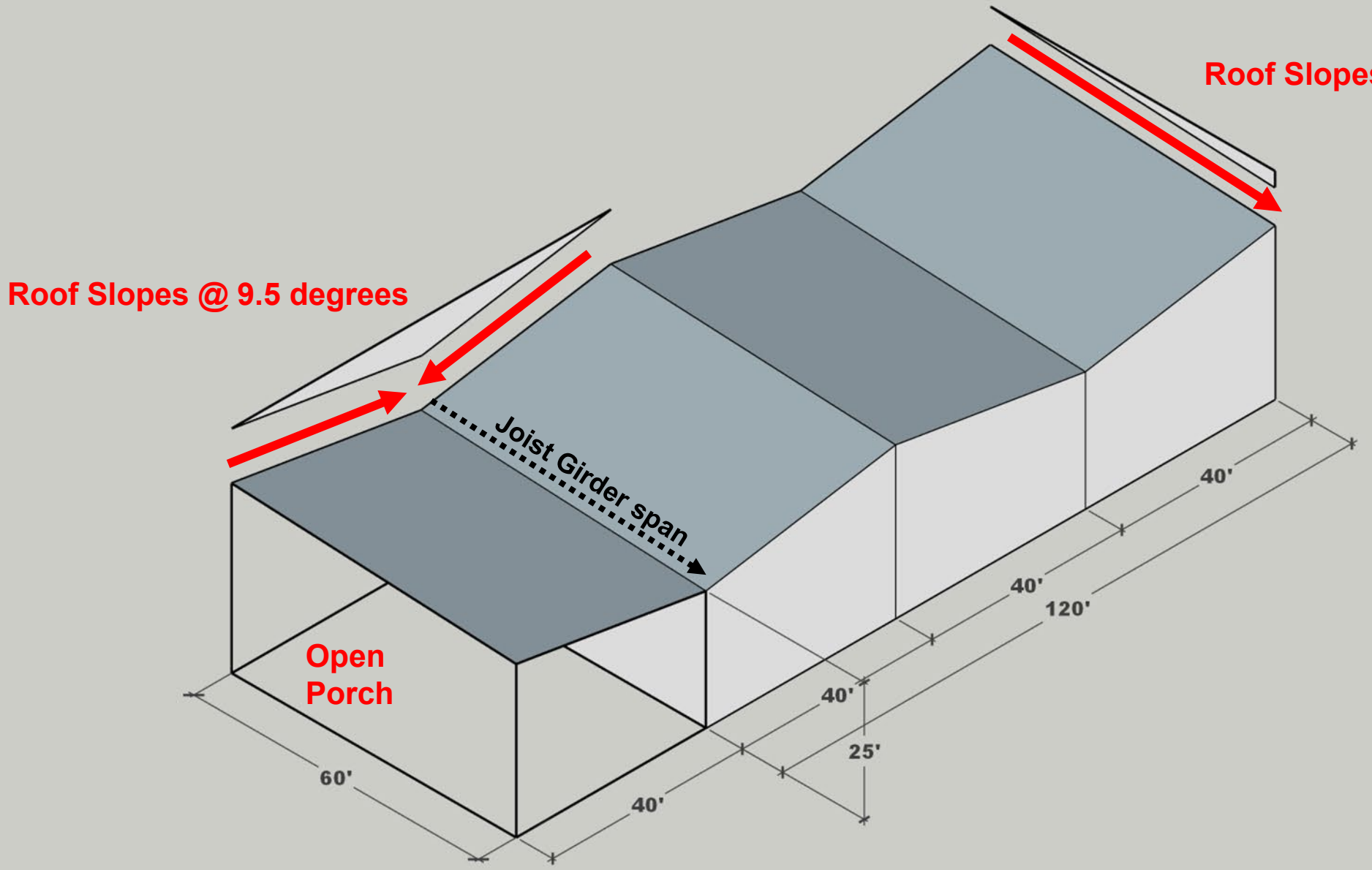
4060 A (40’ x 60’ Bays) **(Open-Web Joists and Joist Girders can work 2 ways)**



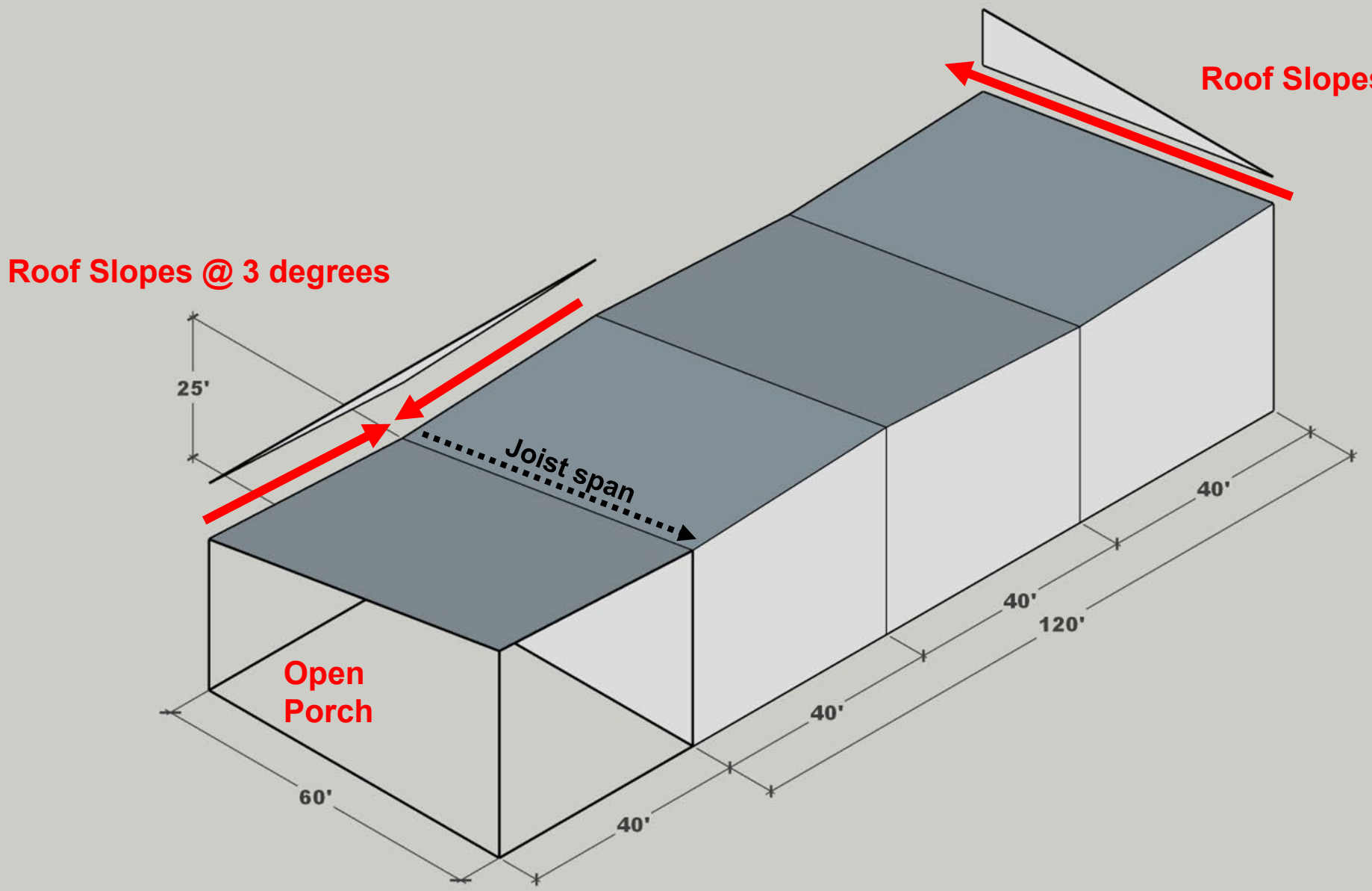
4060 B (40’ x 60’ Bays) **(Open-Web Joists and Joist Girders can work 2 ways)**



4060 C (40’ x 60’ Bays) **(Joists must span 60’)**



4060 D (40’ x 60’ Bays) **(Joist Girders must span 60’)**



4060 E (40’ x 60’ Bays) **(Joists <u>must</u> span 60’)**

STEEL BUILDING ROOF

PROJECT PARAMETERS: Roof & Overall Building Size

ENGINEERING OFFICE: Ypsilanti, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)
Snow Load: 20 psf Roofing DL: 15 psf

GROCERY STORE: Pinconning, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)
Snow Load: 40 psf Roofing DL: 15 psf

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)
Snow Load: 100 psf Roofing DL: 15 psf

STEEL BUILDING project

PROJECT PARAMETERS FOR ROOF

TERTIARY ELEMENTS: (span determined by specific snow load and ASD load calculations)

Type-B structural corrugated steel Roof Deck: 1.5” thick, 36” width (see span & load table)
Roof Deck must have a cantilever beyond Open-Web Steel Joists (only in deck span direction)

SECONDARY ELEMENTS: (spacing determined by corrugated steel deck span)

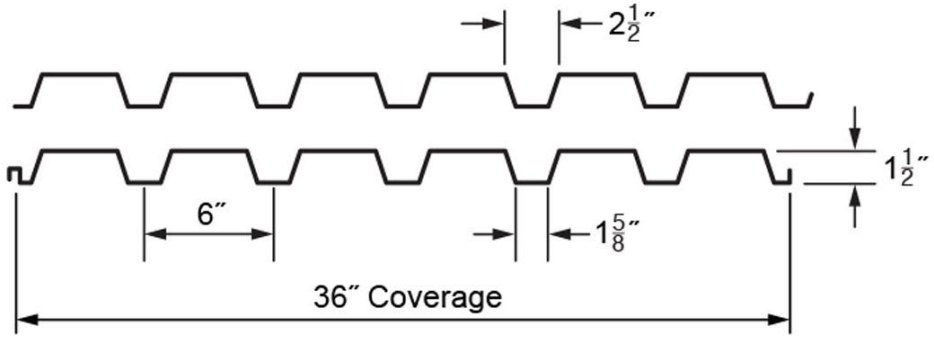
Open-Web Steel Joists: K or LH Series (size and self-weight by economy span & load tables)
Joists must have a top-chord extension for a cantilever beyond building footprint

PRIMARY ELEMENTS: (panel size determined by open-web joist spacing)

Joist Girders: Type-G configuration (size and self-weight by span & load tables for joist loads)
Joist Girders must have a top-chord extension for a cantilever beyond building footprint
Joist Girders supported on steel Columns (Wide-Flange)



B Deck



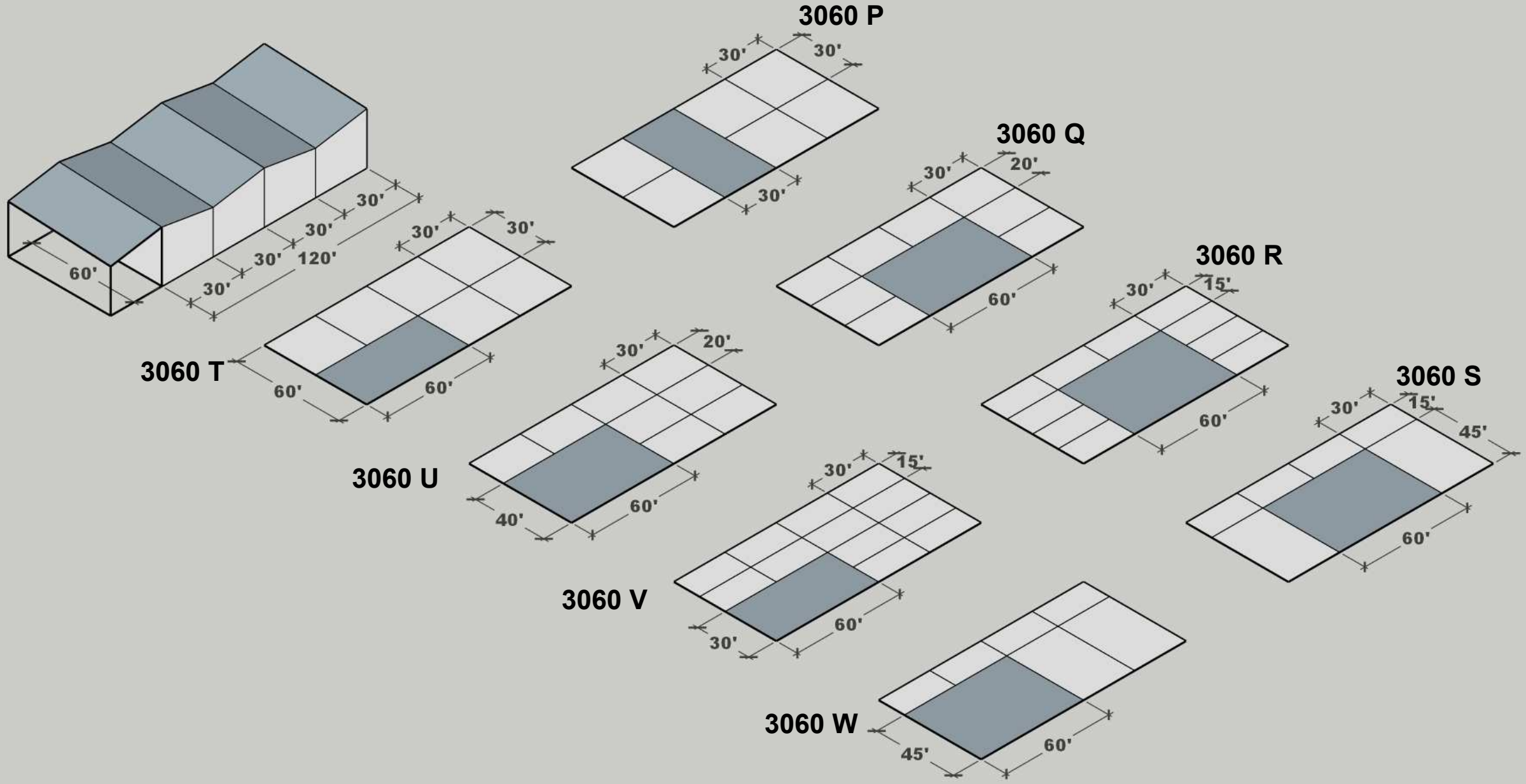
MAXIMUM CANTILEVER SPANS

Gage	F_y (ksi)	Back-Span Condition		
		Single	Double	Triple
20	40	2 - 0	1 - 11	1 - 11

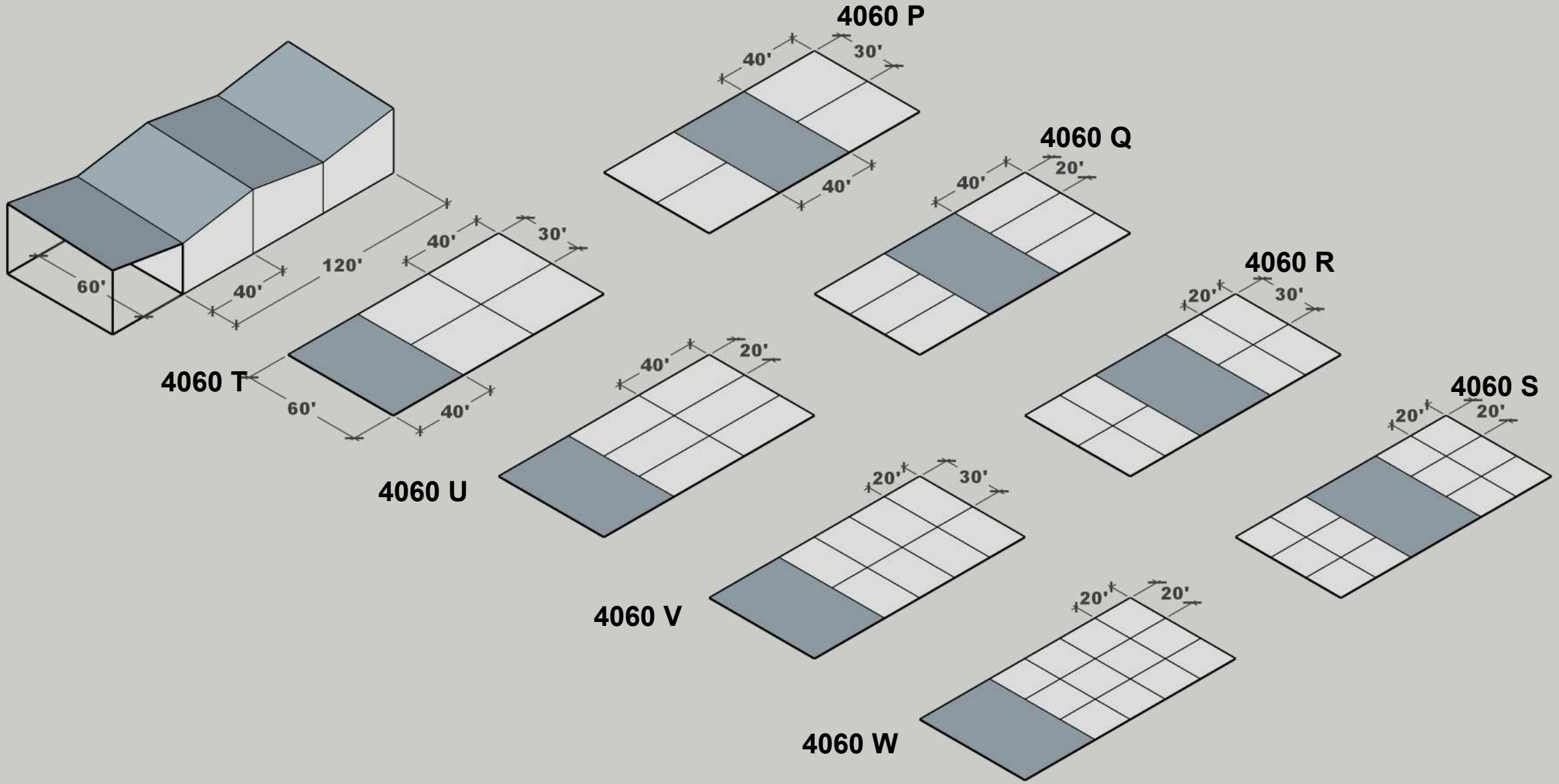
ALLOWABLE UNIFORM LOADS AND MAXIMUM CONSTRUCTION SPANS

Span Condition	Gage	Weight (psf)	Allowable Uniform Total Load (psf) Center to Center Span (ft - in)										SDI Max. Constr. Span (ctr / ctr)
			5 - 0	5 - 6	6 - 0	6 - 6	7 - 0	8 - 0	9 - 0	10 - 0	11 - 0	12 - 0	
Triple	20	1.98	183	152	128	109	94	72	57	47	39	-	10 - 8

ROOF DECK B-Type: Span & Load Table (20 gage, with Triple Span Condition)



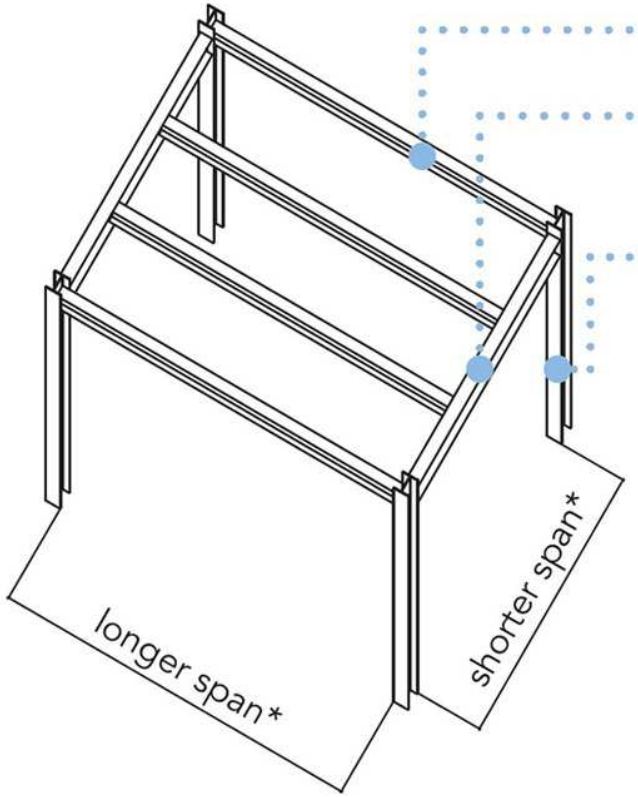
FLOOR LAYOUT OPTIONS: 3060 Second-Floor Plan (blue grey area is open to roof in plan)



FLOOR LAYOUT OPTIONS: 4060 Second-Floor Plan (blue grey area is open to roof in plan)

STRUCTURAL STEEL DESIGN TIPS

Quick Estimates and Rules of Thumb for Architecture Students



Beam (framing to transfer floor load to girder)

Girder (a beam that transfers the load of several beams to columns)

Column (vertical support that transfers girder loads to the foundation)

*The most efficient and economical bays have a 1:1.25–1.5 width proportion and an area of 750–1,250 ft² with infill beams spanning the long direction to have closer girder and beam depths. Spacing should consider the type of deck and spans.

RECOMMENDED BAY RATIOS & GIRDER/BEAM SPAN ORIENTATIONS

STEEL BUILDING FLOOR

PROJECT PARAMETERS: Floor

ENGINEERING OFFICE: Ypsilanti, Michigan
Floor LL: 65 psf Floor DL: 15 psf
Second-Floor: 4,200 sf to 5,400 sf (see layout options)
1.5 CD Composite Deck with concrete: 33 psf (31+2)

GROCERY STORE: Pinconning, Michigan
Floor LL: 75 psf Floor DL: 15 psf
Second-Floor: 4,200 sf to 5,400 sf (see layout options)
1.5 CD Composite Deck with concrete: 33 psf (31+2)

PUBLIC LIBRARY: Houghton, Michigan
Floor LL: 150 psf Floor DL: 15 psf
Second-Floor: 4,200 sf to 5,400 sf (see layout options)
1.5 CD Composite Deck with concrete: 33 psf (31+2)

STEEL BUILDING FLOOR

PROJECT PARAMETERS: Floor

Tertiary Elements (span determined by floor load)

1.5CD Composite Deck: 3.5” total depth/ deck thickness, 36” width (see span & load table)

Secondary Elements (spacing based on deck span; span on layout, sizing on load)

Wide-Flange Steel Beams (size using Tributary Load and OCHSHORN Steel Beam Calculator)
Lateral Load-Resistance Diagonal Bracing: Tension Rods, WF section, HSS Square Tube, or Angle

Primary Elements (Girder span based on layout, sizing on load) (Column sizes based on load and un-braced height)

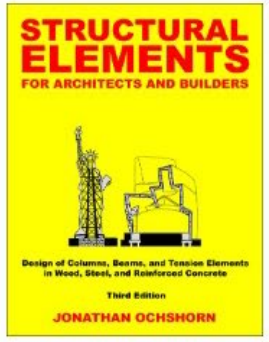
Wide-Flange Steel Girders (size using Tributary Load and OCHSHORN Steel Beam Calculator)
Wide-Flange Steel Columns (size with OCHSHORN Steel Column Calculator after Beam & Girder tributary load determined)

STEEL BUILDING project

PROJECT CALCULATORS: Steel Beams (Girders & Beams) **and Columns**

OCHSHORN Steel Beam Calculator: https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

OCHSHORN Steel Column Calculator: https://jonochshorn.com/scholarship/calculators-st/steel-column-design/index.html



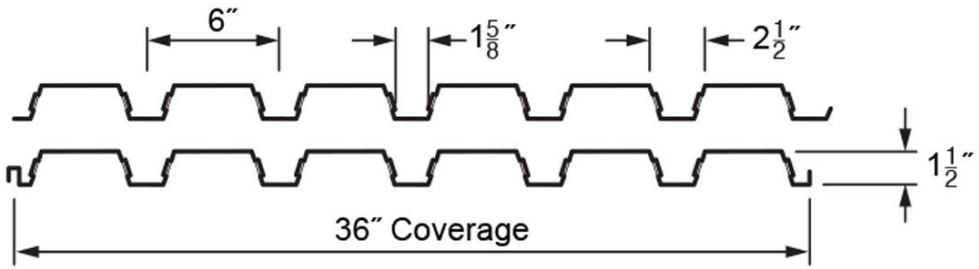
Laterally-Supported Steel Floor Beam Calculator

Jonathan Ochshorn

© 2009 Jonathan Ochshorn.

1.5CD Composite Deck

NORMAL-WEIGHT CONCRETE



SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS

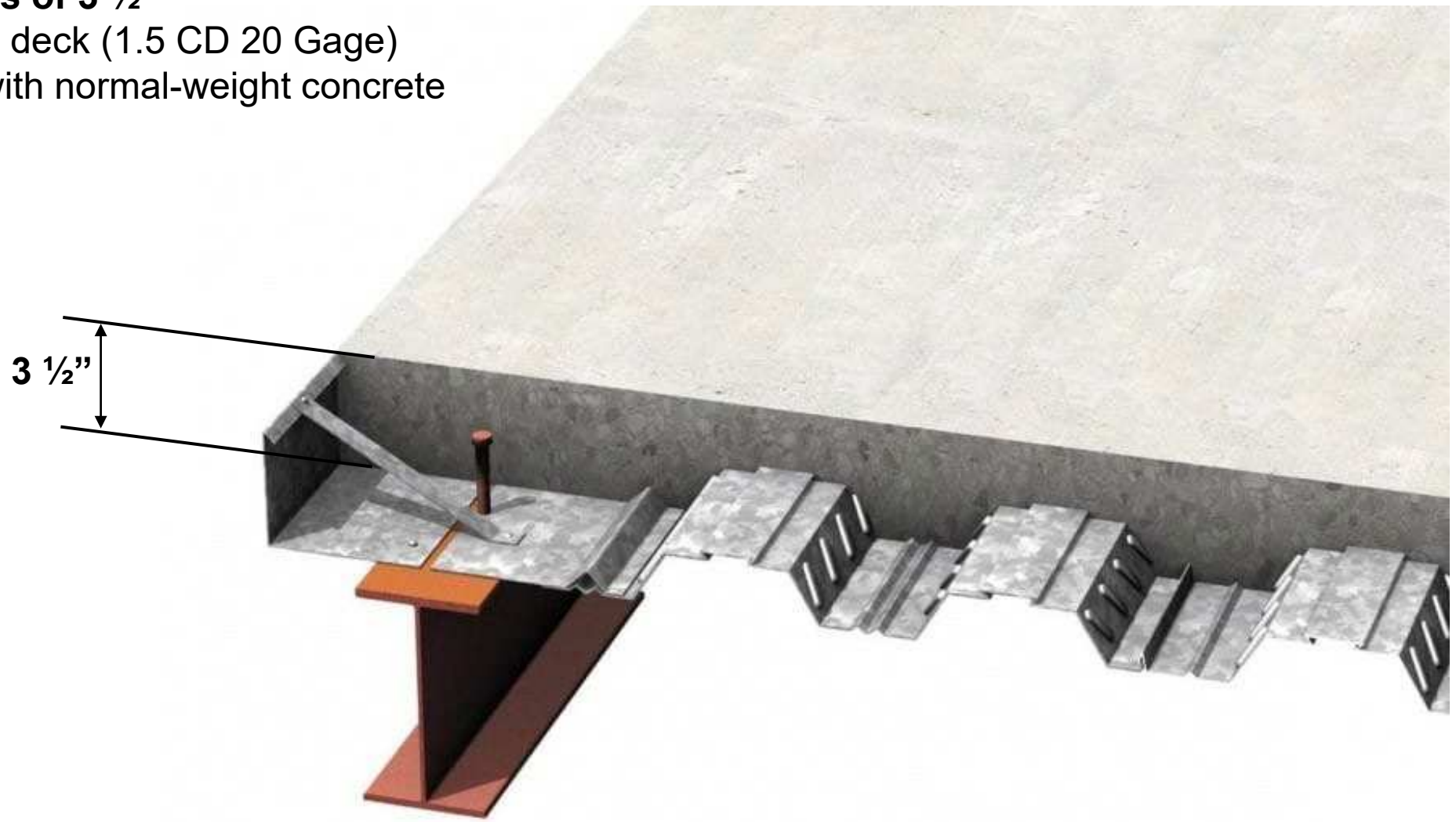
Total Slab Depth (in)	Concrete Weight (psf)		Weight (psf)	Gage	Allowable Superimposed Uniform Load (psf)								
					Clear Span (ft - in)								
					6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0
3 1/2		31	1.98	22	294	246	207	176	151	130	113	85	65
				20	354	297	251	215	185	160	139	106	82
				18	400	386	328	281	243	212	185	139	105
				16	400	400	400	349	301	251	212	154	116

SPAN & LOAD TABLE: Composite Steel Floor Deck (use 20 Gage, with 3 ½ Total Slab Depth)

Floor Deck thickness of 3 ½”

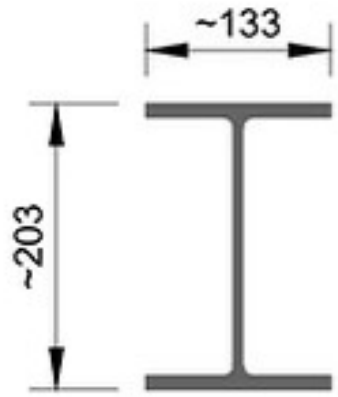
1 ½ “ steel composite deck (1.5 CD 20 Gage)

3 ½” total thickness with normal-weight concrete



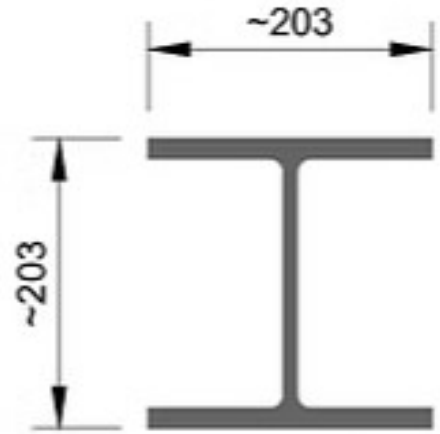
Composite Steel Deck for Steel Building project

Universal Beam [UB]



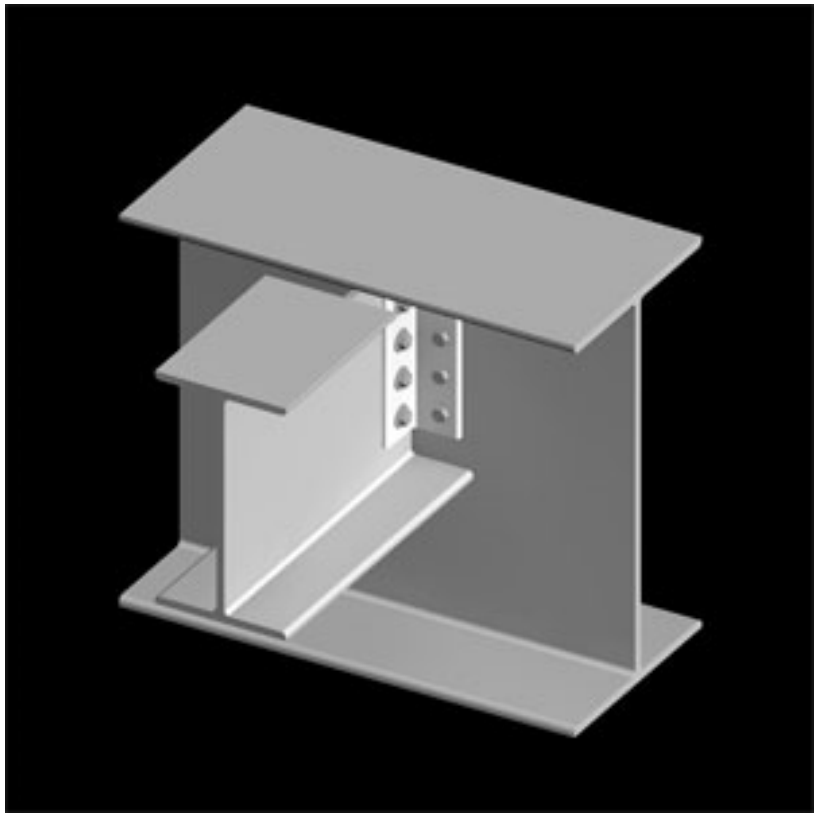
eg. 203 x 133 x 30 UB
UB's are always taller than they are wide

Universal Column [UC]

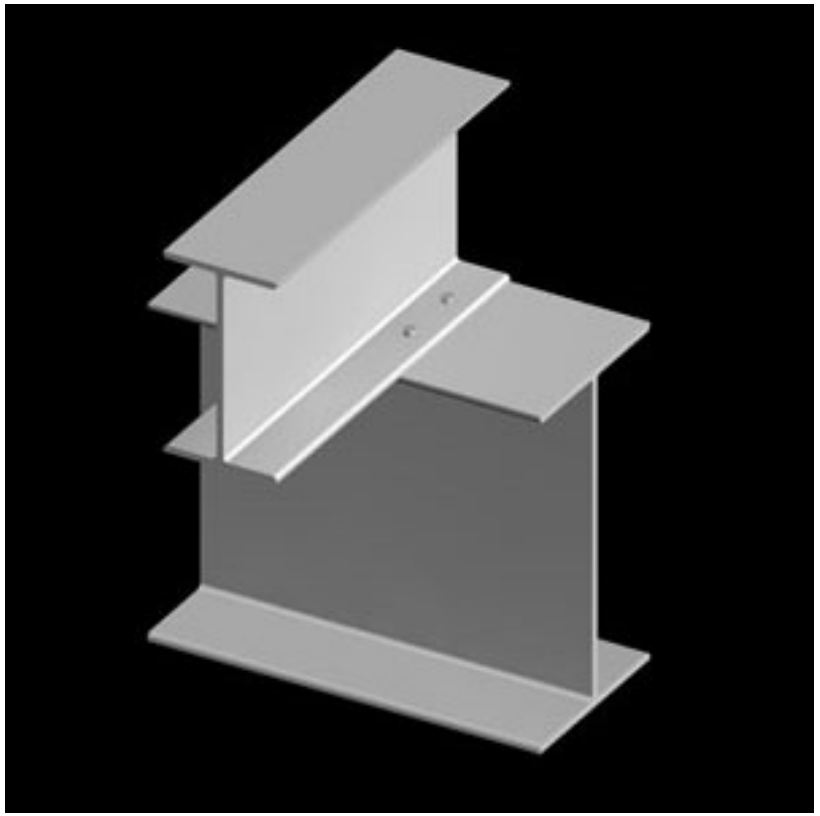


eg. 203 x 203 x 46 UC
UC's are approximately equal in height and width

Wide Flange (WF) as Floor Beam/Girder, and as Columns, for the Steel Building project

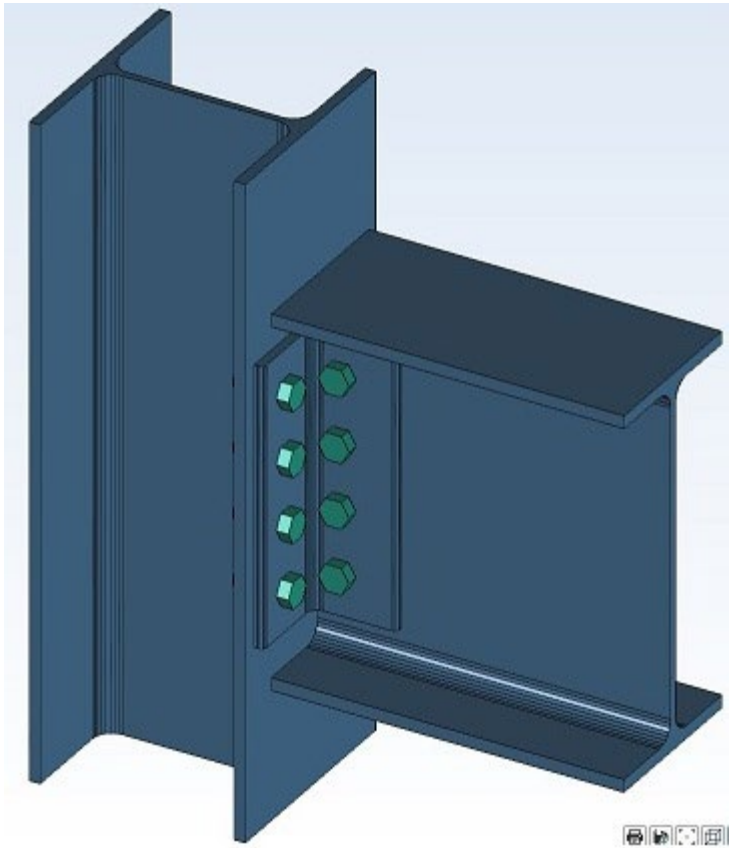


Top Flange framing

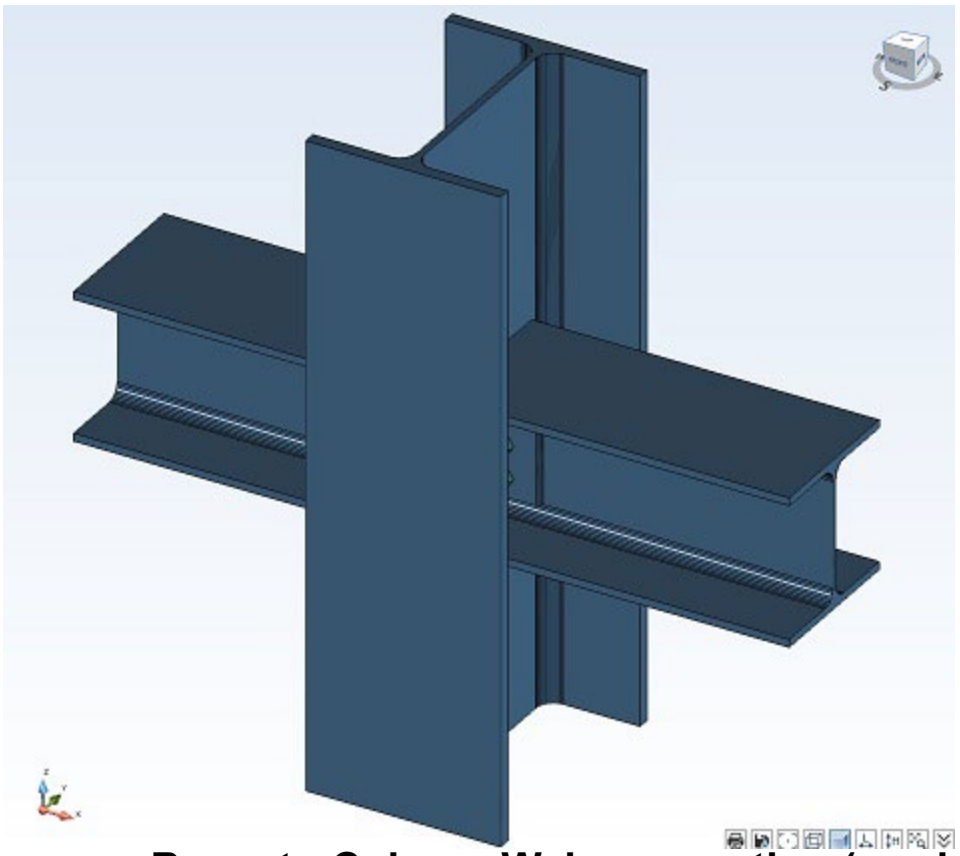


Beam on Girder

Wide Flange (WF) Beam to Girder connections*…choose one type for your floor structure*

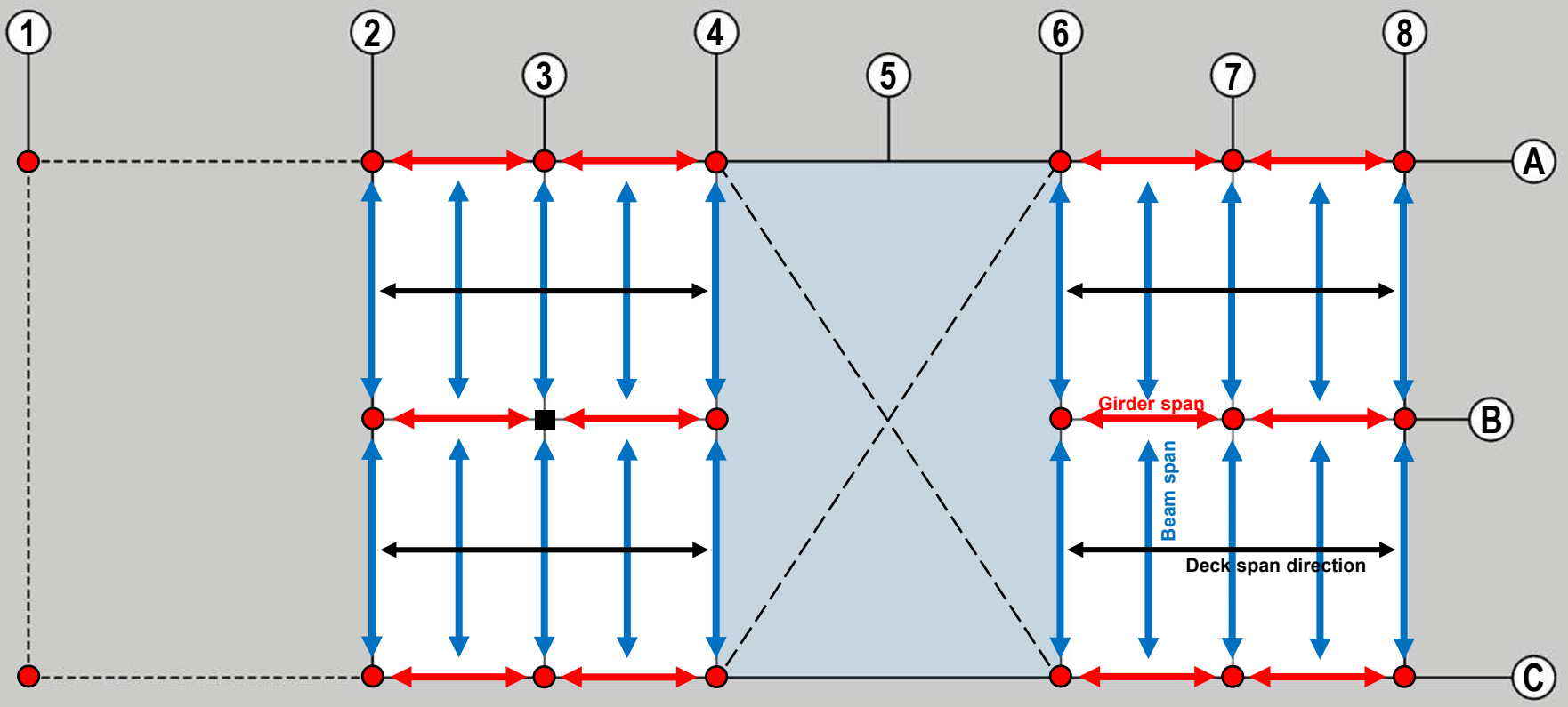


Girder to Column Flange connection (strong axis)

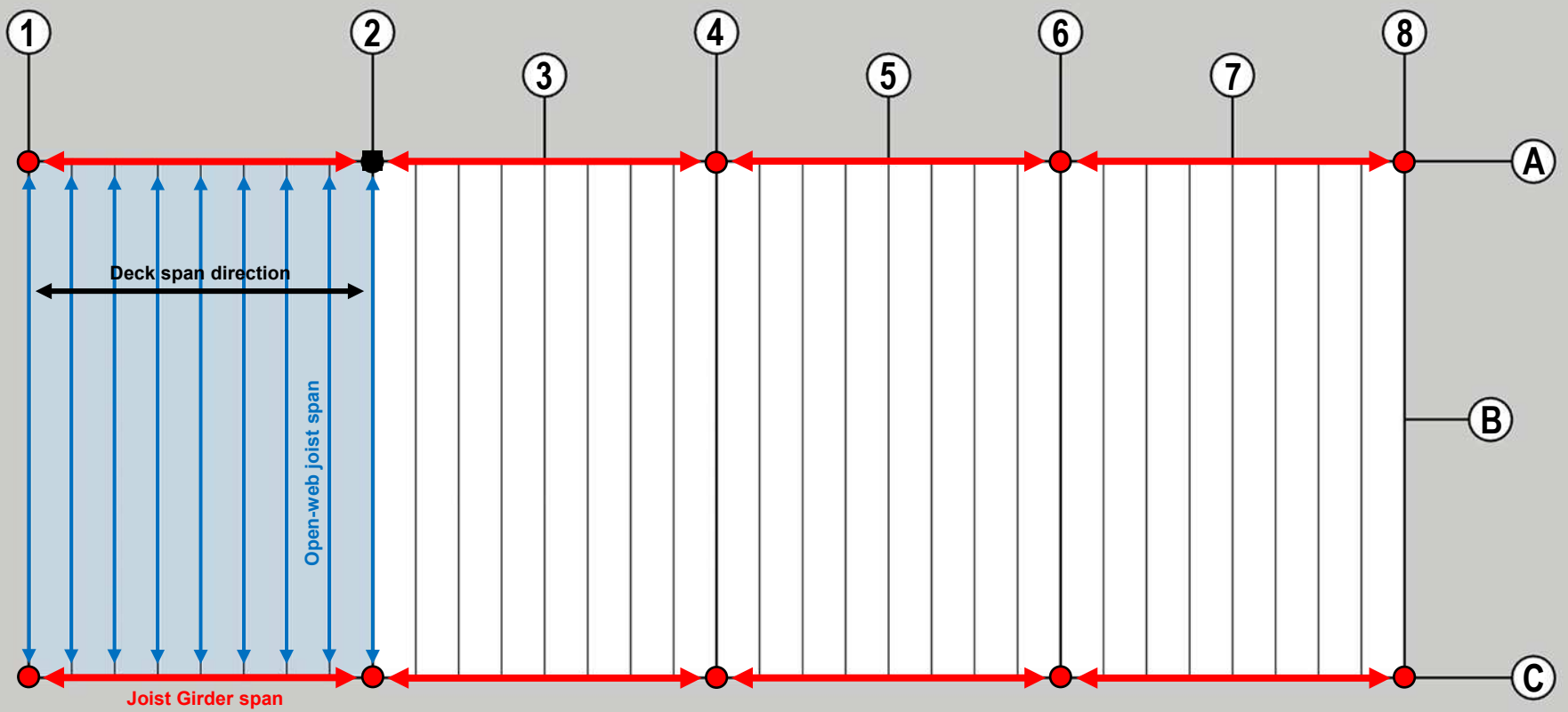


Beam to Column Web connection (weak axis)

WF Beam or Girder to Column connections, for Steel Building project



EXAMPLE: Framing Plan Diagram of floor structure



EXAMPLE: Framing Plan Diagram of roof structure

STEEL BUILDING LATERAL LOAD-RESISTANT MECHANISMS

PROJECT PARAMETERS: Lateral Load-Resistant Mechanisms

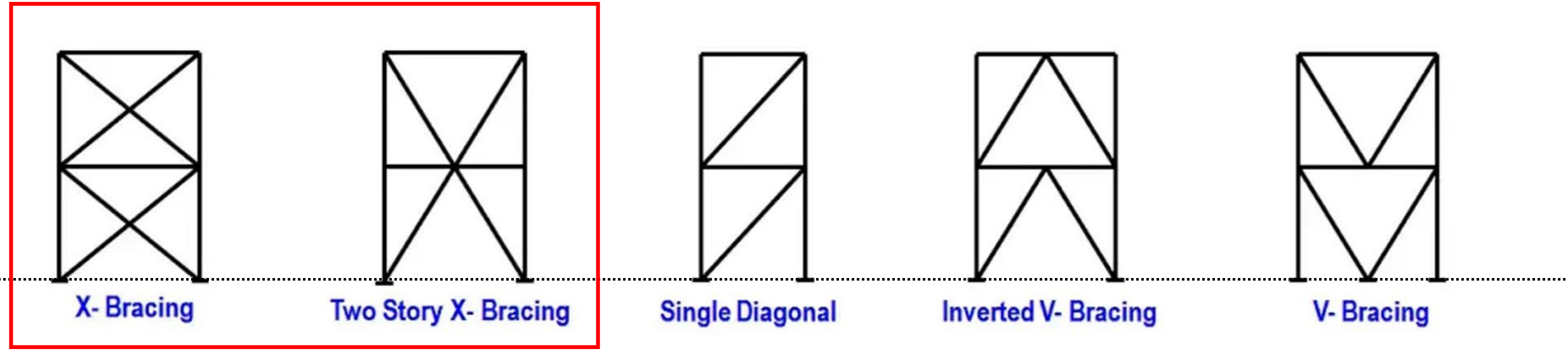
Horizontal Mechanisms (provided by 1.5 roof deck and 1.5 CD composite floor deck)

Diaphragm at Roof Deck & Floor Deck

Vertical Mechanisms (provided by selection of one or more triangulated concentric braced frame options)

Concentric Braced Frames: Tension Rods (or) WF section (or) HSS Square Tube (or) Angle

Note: Minimum of two braced frames in transverse direction & two in longitudinal direction on all four sides of the building.



Concentric Braced Frames transfer lateral forces by connecting to columns and beams at their joints, forming a triangulated "truss." They are called concentric because the axes of the braces, columns, and beams align at the connections. They provide high stiffness and strength to resist lateral loads.

…any one of these geometries can be applied to our Steel Building.
…it must have triangulation.

CONCENTRIC BRACED FRAMES, ***that can be applied to the Steel Building project***

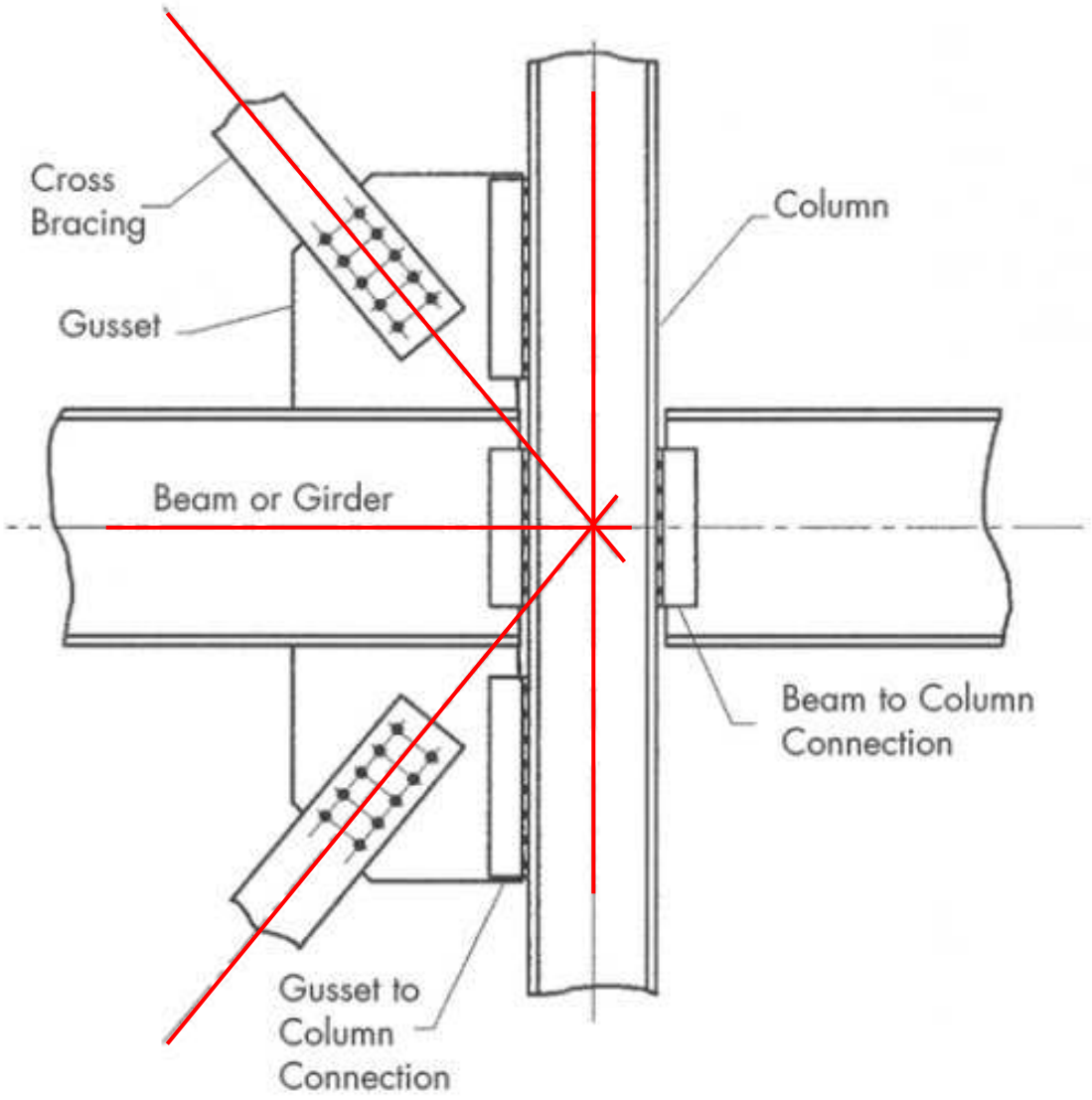
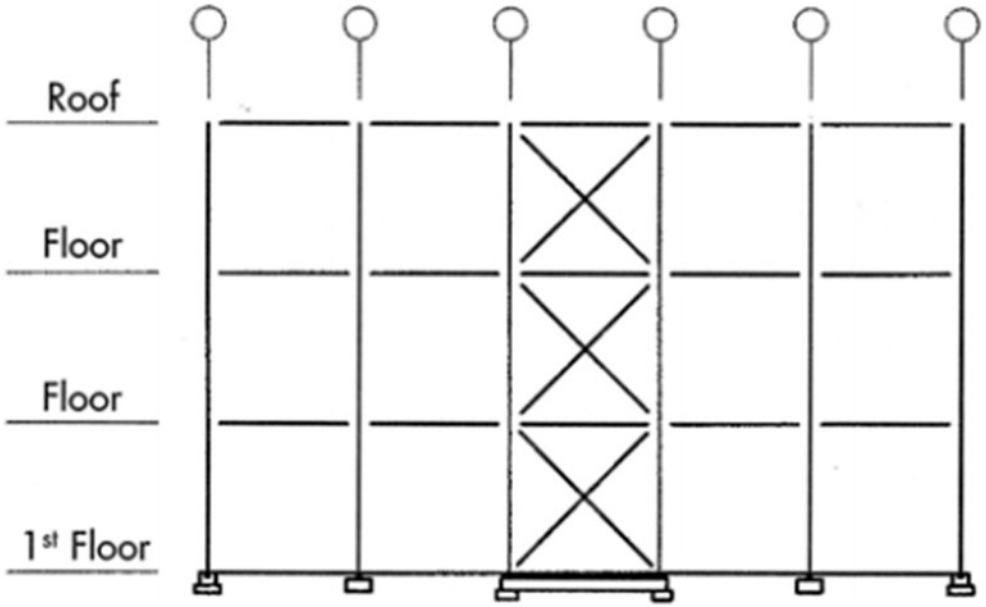
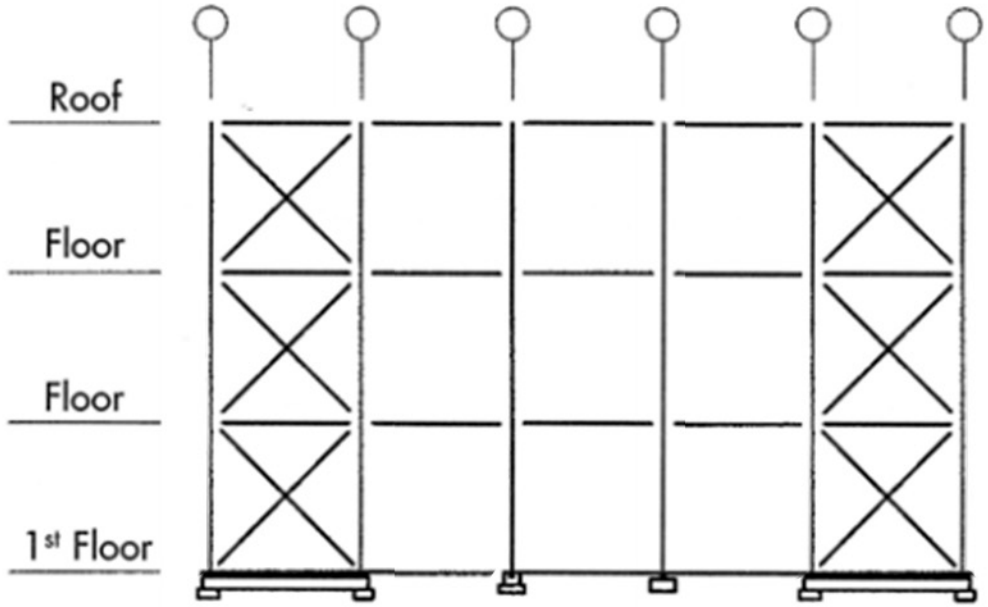


Fig. 6. Typical beam to column brace connections.

CONCENTRIC BRACED FRAMES axis always align at a concentric point (like truss nodes)



Across the central bay.
Four braced planes: two transverse and two longitudinal



At the four corners.
Four braced planes: two transverse and two longitudinal

CONCENTRIC BRACED FRAMES common application options

STEEL BUILDING FOUNDATION

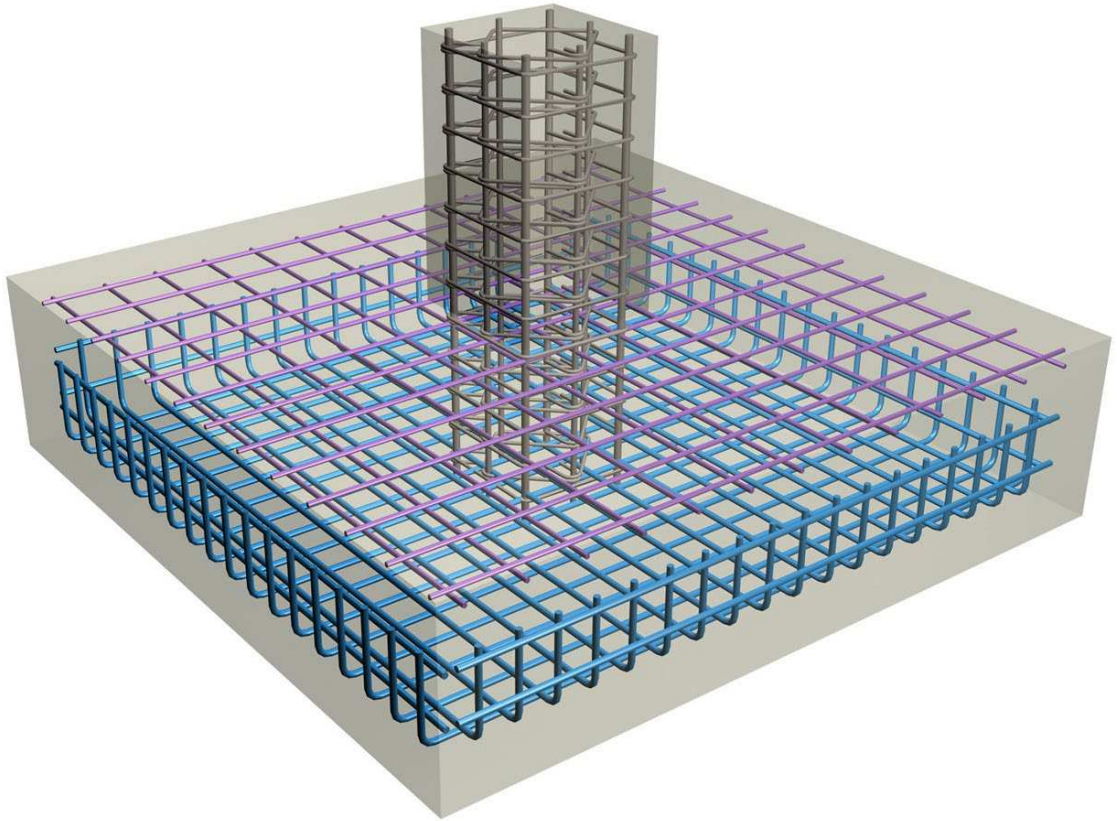
PROJECT PARAMETERS: Foundation

Foundation (for all projects)

Spread footing with bottom at frost line, and 20” x 20” concrete pedestal on footing, extending up to grade (bottom of steel column)

ENGINEERING OFFICE: Ypsilanti, Michigan	Frost Line: 42” below grade
GROCERY STORE: Pinconning, Michigan	Frost Line: 42” below grade
PUBLIC LIBRARY: Houghton, Michigan	Frost Line: 60” below grade





TYPICAL SPREAD FOOTING WITH CONCRETE PEDESTAL FOR STEEL COLUMN

STEEL FLOOR

SECONDARY HORIZONTAL STRUCTURE

BEAM SPACING

Beams are spaced at 6.67’ o.c. to optimize the floor deck to span as close to 7’ as possible, while conforming to the 20’ x 30’ structural bay for the floor system. The 20 x 30 bay is the preferred bay ratio of 1.5 to 1, with Beams spanning in the 30’ direction and Girders spanning 20’.

TRIBUTARY LOAD ON BEAM A

Composite Deck Floor Load: 198 psf
Deck span and Beam spacing: 6.67’
Uniformly Distributed Tributary Load on Beam calculation: 198 psf (6.67’) = 1,321 plf
LOAD ON BEAM A: 1,321 plf

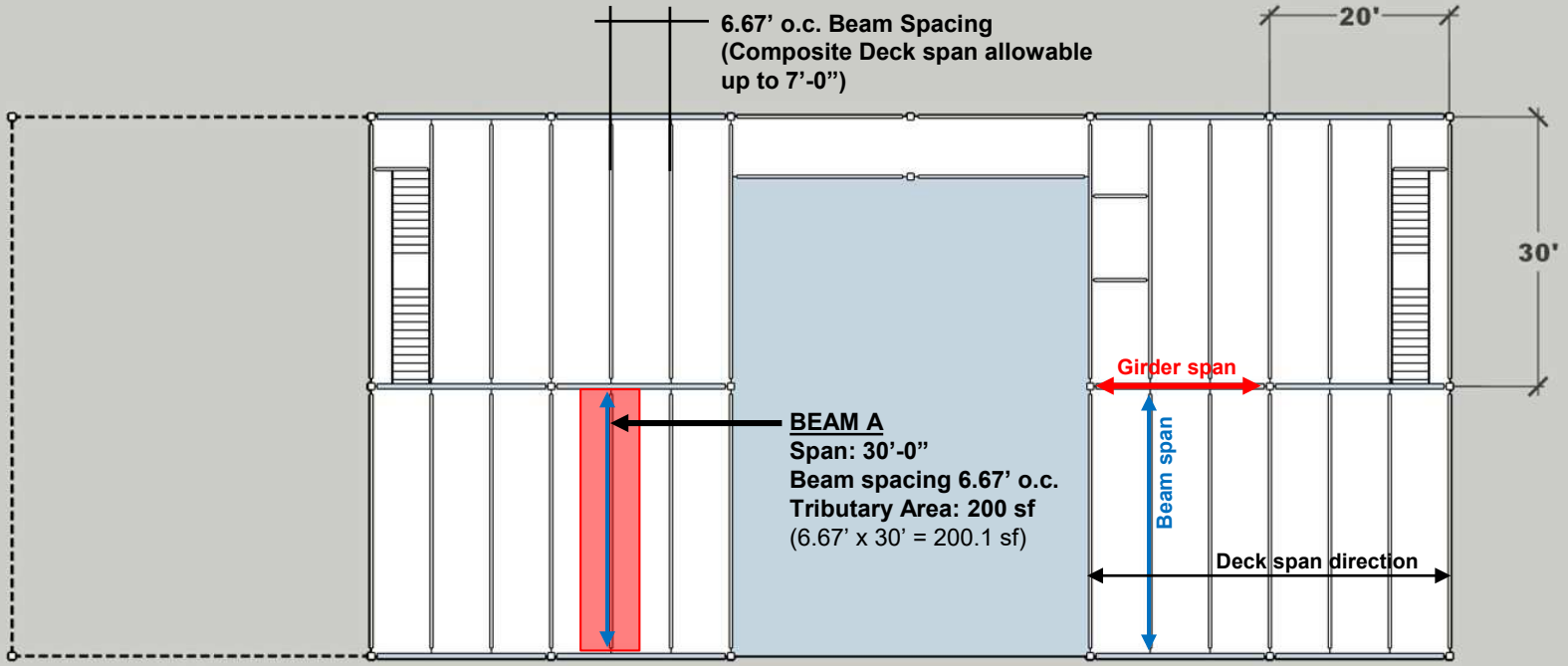
SIZING OF BEAM A

Sizing based on values entered and results indicated using the following online structural Steel Beam Calculator:
https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

Results indicated a wide-flange beam W18 x 35 is the lightest section to meet the load and beam span criteria. This beam is 17.7 inches deep and weighs 35 plf.

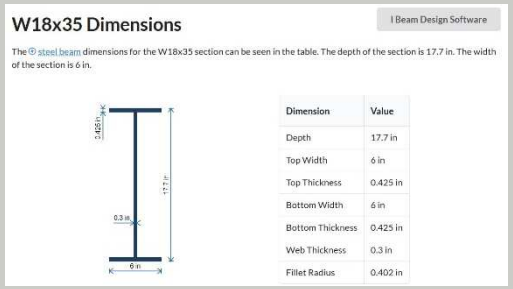
SELECTION FOR BEAM A: W18 X 35

SAMPLE FORMAT!!!

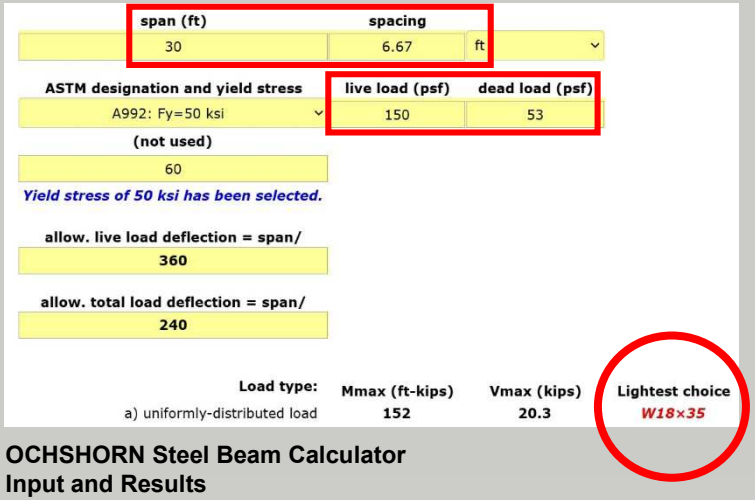


BEAM A

Total Tributary Load: 39.62 kips
(198 psf x 200.1 sf)
Uniform Distributed Load: 1,321 plf
(198 psf x 6.67’ = 1,321 plf)



W18X35 size



OCHSHORN Steel Beam Calculator Input and Results